

# 2023 Ch H1 Q3

Section: Nature's Chemistry

Topic: Fats and Oils

## Question Summary

Fats are formed from glycerol molecules and fatty acid molecules. What is the mole ratio of glycerol to fatty acid molecules?

A: 1 : 2

B: 2 : 1

C: 1 : 3

D: 3 : 1

## Worked Solution

A triglyceride (fat/oil) molecule is formed by the condensation of one glycerol molecule with three fatty acid molecules. Each of the three  $\text{-OH}$  groups on glycerol reacts with the  $\text{-COOH}$  group of a fatty acid to form an ester link.

## Final Answer

**C — 1 : 3**

## Revision Tips

- Glycerol has three hydroxyl (-OH) groups.
- Each can form an ester link with a fatty acid's -COOH group.
- Therefore, 1 glycerol + 3 fatty acids → triglyceride.